

Introduction

What is a breadboard?

The breadboard or solderless breadboard is the most popular tool used for prototyping (building temporary electronic circuits and testing them before the final version is permanently designed) without any need for soldering.

You can use it as an independent circuit to create your project, or it can be used as an auxiliary part of a main circuit such as Arduino or Raspberry pi.

The breadboard has a user-friendly construction and quite a safe layout, which make it an important step for newcomers to electronics who wish to test their ideas in quick experiments. However, you can use the breadboard to build all types of electronics projects, starting from very basic circuits to the most complicated robotics.

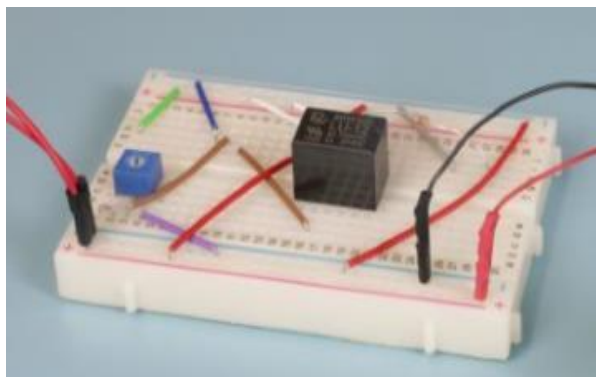
What's the reason behind the name "breadboard"?

You may ask "what does a breadboard have to do with electronics?" And if your imagination pictures a fresh loaf on a piece of wood, then you are not far from guessing where the name comes from.

The name "breadboard" goes way back to the days when people would literally use a piece of wood on which they sliced bread, in order to construct electronic circuits. They used nails and wires to connect up the components. Nowadays, the new generation of "breadboard" is made from a rectangular piece of plastic, but we still refer to them by the name "breadboard."



Breadboard circuit in the past



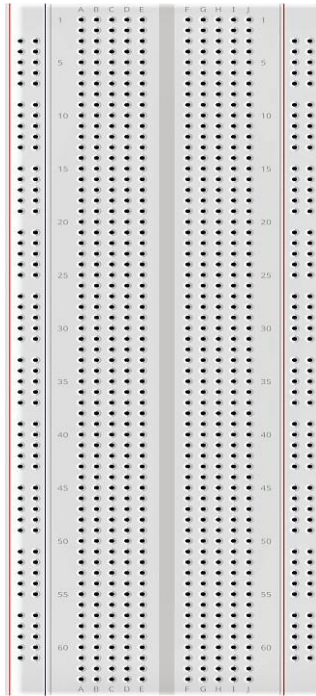
Breadboard circuit today

Types of breadboard

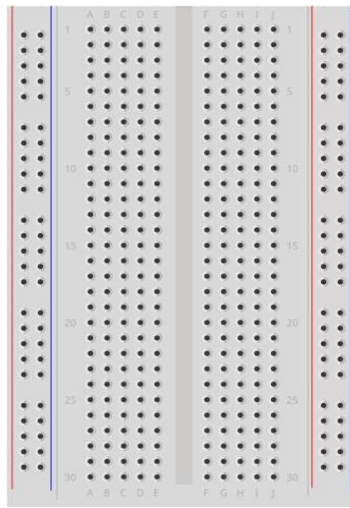
Breadboards come in various brands and sizes to suit the requirements of the electronic circuit. Nevertheless, all breadboards share the same fundamentals and ideas.

Here are some types of a standard breadboard:

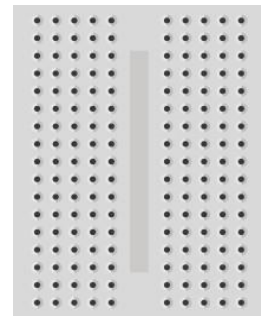
1. Full-size
2. Half-size
3. Mini



Full-size

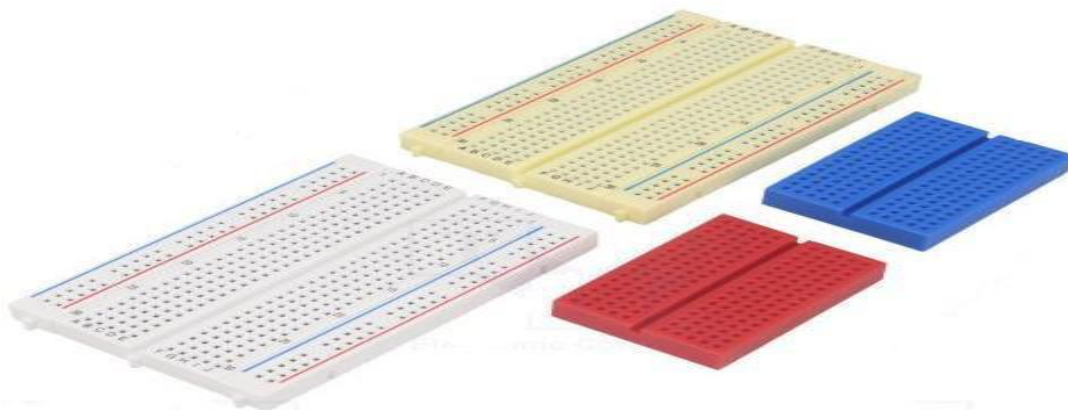


Half-size



Mini

Moreover, there are different colors of breadboards (white, blue, red, green, yellow, transparent).



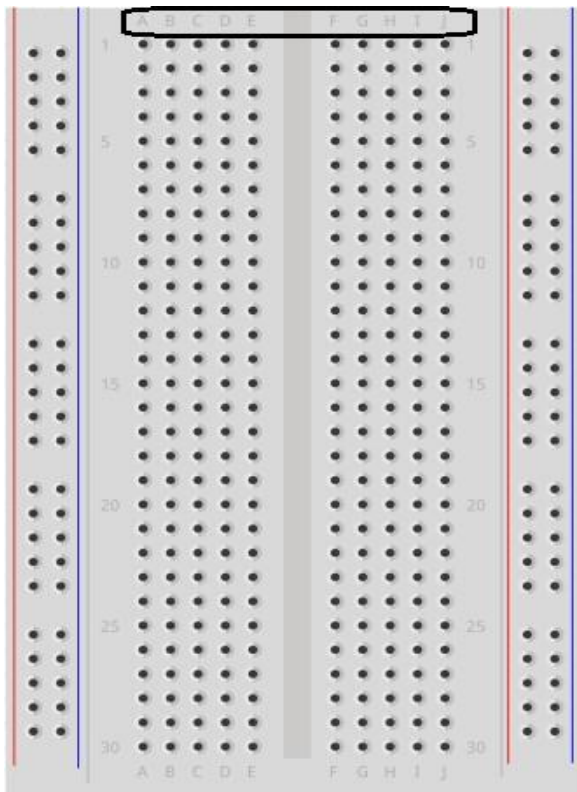
Breadboard layout

What is the breadboard terminology?

Almost all modern (full-size and half-size) breadboards have numerous holes, letters, numbers, and blue and red lines that give the breadboard its standard appearance. Here is the terminology you must know:

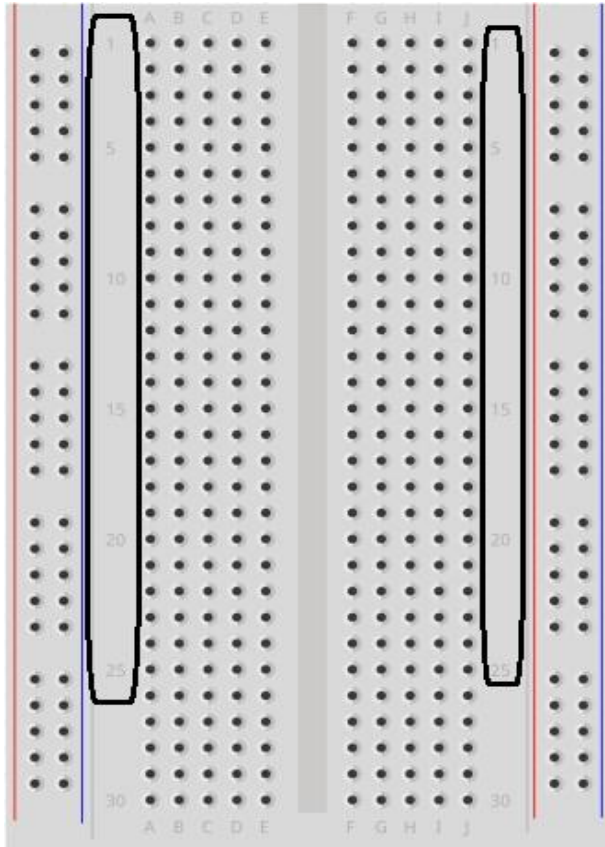
Letters

Printed at the top and bottom of the breadboard are letters from “A” to “J” running horizontally and dividing the holes evenly into vertical lines.

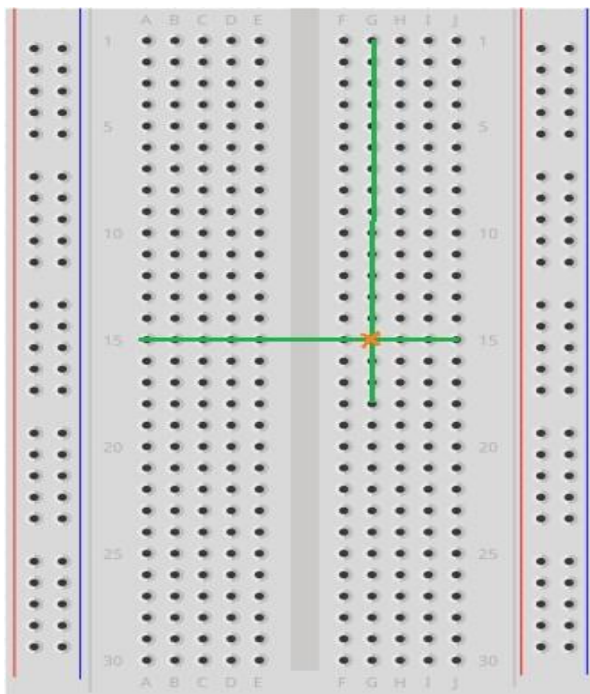


Numbers

To identify horizontal rows, numbers from 1 to 30 are printed on left and right edges on half-size breadboards, and to 60 or 63 on full-size breadboards. Some breadboards use skip-counting in fives, but it doesn't change the main concept of row-numbering on breadboards.



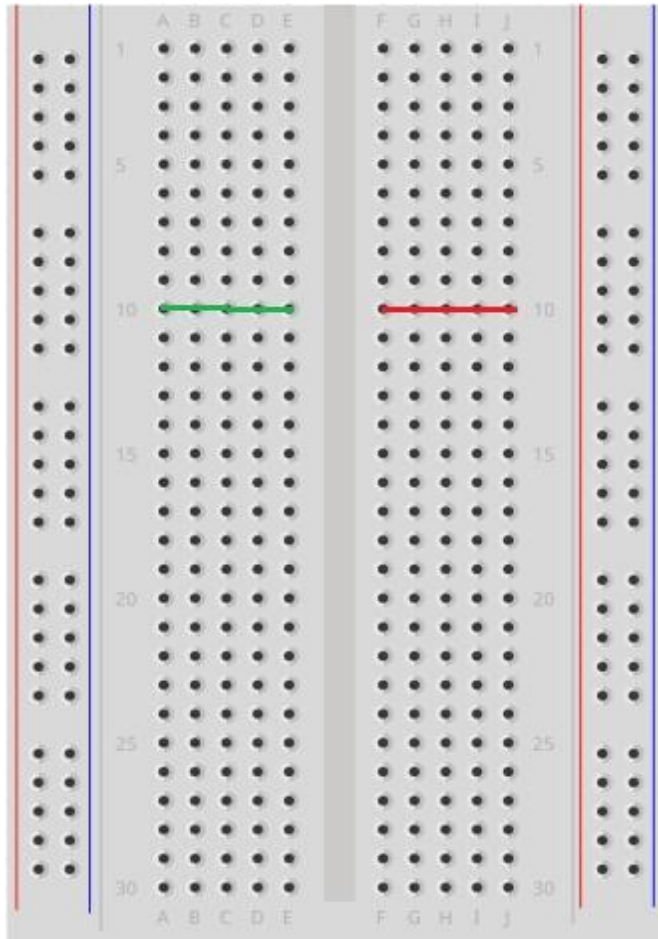
Together, the numbers and letters guide you to locate a certain hole easily while building complicated circuits or during troubleshooting. For instance, hole “G15” is located at the intersection of column “G” and row “15.”



Horizontal Row

The rows or “Terminal strips” are where you are supposed to connect the electronic components. Rows are identified by numbers, and each row contains five holes on each side (left and right) separated by a middle divider. Every five holes in the same row are electrically connected, but they must be on the same side.

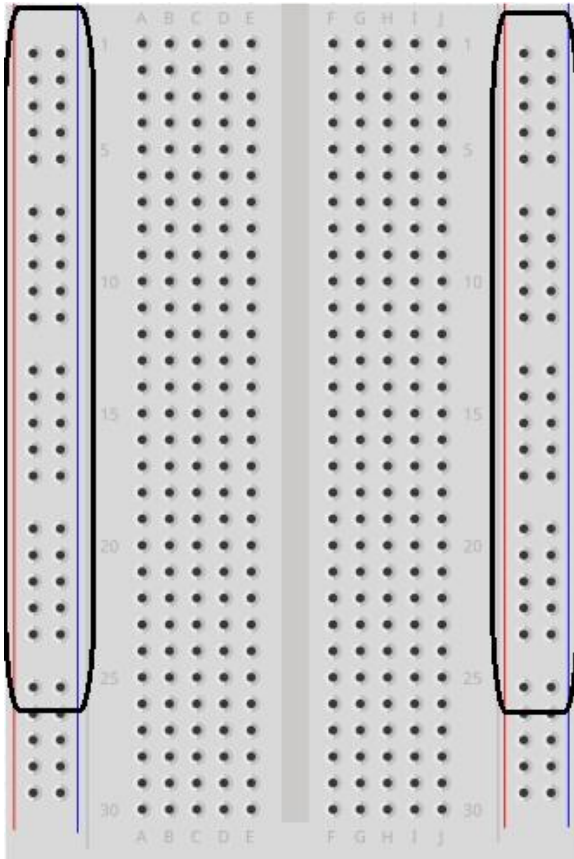
For example, in a row no. 10 the holes 10A, 10B, 10C, 10D and 10E are connected together, because they are in the same row on the same side. But they are not connected to holes F to J on the right side of the breadboard, because both sides are isolated by the middle divider.



Vertical bus

This consists of two vertical columns that are commonly known as “Power rails.” Each breadboard contains two power rails located at the ends of both sides on the breadboard.

The purpose of the power rails is to connect the circuit to an external power supply. One column is to connect the circuit with the positive voltage (+) and the other column is to connect the circuit to ground (-).



How to distinguish between positive voltage column and ground?

Positive voltage column:

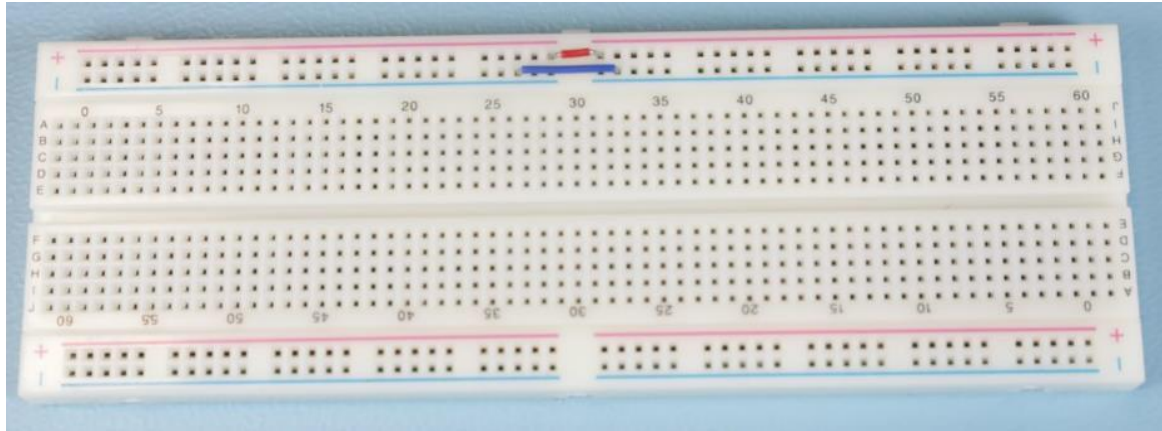
- Located next to the red line
- Marked by a plus (+) sign (*optional!*)

Ground column:

- Located next to the blue or black line
- Marked by a minus (-) sign (*optional!*)

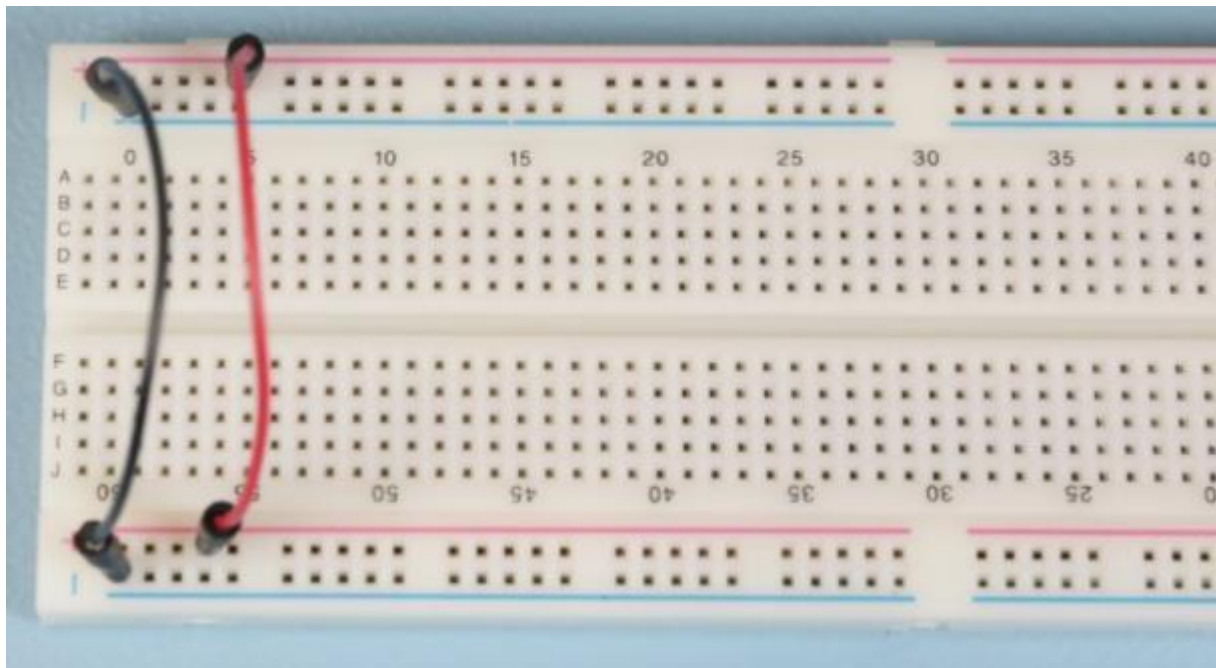
Note that breadboards are available from different manufacturers. Some breadboards use colored lines to identify power rails, other breadboards use colored lines with plus (+) and minus (-) signs.

Tips! It's good to notice that in some full-size breadboards the red and blue (or black) lines are broken in half, which means the upper half of the power rails is electrically isolated from the lower half to enable the use of one breadboard to connect two different circuits. If you wish to connect the two halves together, it's enough to use jumper wires as shown below.



Two halves of the breadboard are connected via red and blue jumper wires

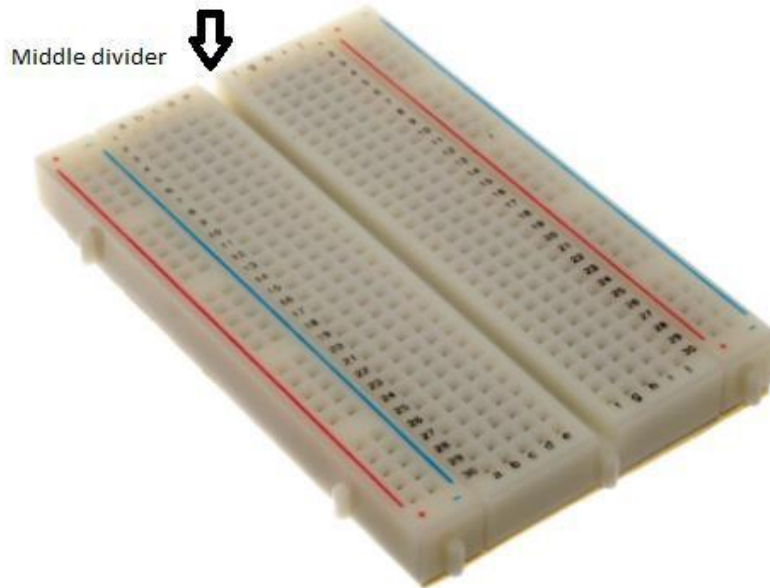
Tips! The power rails on the left side are isolated from the second set of power rails on the right side. Use jumper wires if you want positive voltage and ground to be available on both sides. (Remember to connect positive to positive and ground to ground.)



Power rails from both sides are connected to each other via jumper wires

Central gutter

A middle divider or “ravine” that runs vertically and divides the breadboard symmetrically into left and right sides. As we mentioned earlier, the halves of the breadboard are electrically isolated from each other by the middle divider.



This ravine has very important functions:

1. It allows dual-in-line packages (DIPs) such as ICs to fit properly on the breadboard. As each pin of an IC serves a unique purpose, this ravine keeps the pins isolated from each other to avoid short circuits.
2. It provides cooling for DIP components by allowing air to flow from beneath. ICs chip get warm under some circumstances.

Tips! Use an IC chip lifter or tweezers to remove ICs from the breadboard easily and safely.

How to use the breadboard?

What's inside the breadboard?

If your curiosity drives you to check what's inside the breadboard, you will see rows of conductive metallic clips under the holes.

The clips have a spring-loaded feature that holds the leads of components firmly when you insert them into the holes.

Remember, every set of five holes in a row are electrically connected, because each metallic row contains five clips. Under each row of five holes there is a metallic row of five clips: one clip under each hole! That means if you insert any component into one of the five clips, the rest of the clip's holes in the same row will be electrically connected to this component.

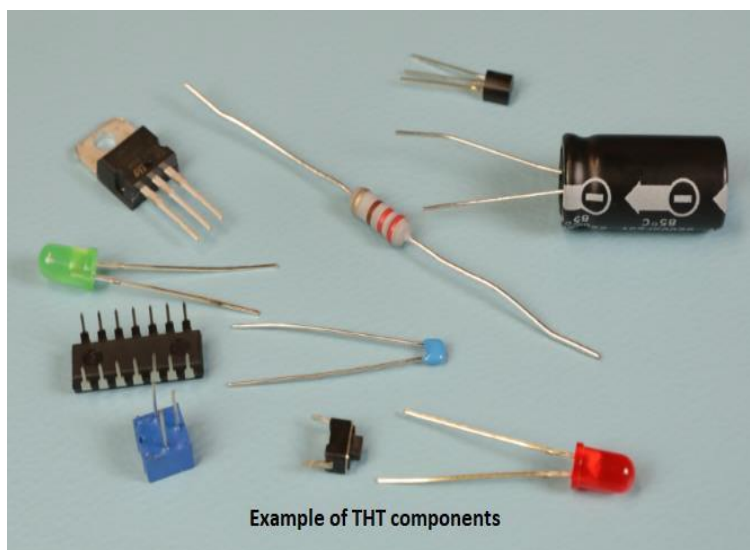
The power rails consist either of one long metal bar with many clips or two isolated bars in the case of breadboards where the upper half is isolated from the lower half.

Tips! As we mentioned, every set of five holes in the row are electrically connected by default; however, you can always connect other holes on the breadboard together by using a jumper wire, for example, or any other component such as a resistor.

Which components are best to use with breadboards?

Having explained how breadboards work from the inside, it's easy to guess which components are compatible with the breadboard.

Almost all components that have leads and are known as THT (through the hole technology) components can be used on breadboards. Since there is no need for soldering, we can connect the components simply by pushing the leads into the hole's metal clips.



Tips! Some components do not have leads, such as variable resistors and DC motors, but you can still use them on the breadboard if you solder a wire to their terminals. Then you can use the wires as leads to be pushed into the holes.